

20. What is the probability of drawing 2 vowels and 2 consonants, in any order?
21. What is the probability of drawing at least 1 vowel when drawing four tiles? (Hint: use the Formula for Complements.)
22. What is the probability of drawing at least 1 consonant when drawing four tiles?

The following exercises involve the board game *Clue*, which involves a deck of 21 cards: 6 suspects, 6 weapons, and 9 rooms. At the beginning of the game, 1 card of each of the 3 types is secretly removed from the deck (the object of the game is to identify those 3 cards). The remaining 18 cards are dealt out to the players. Assuming there are 3 players, each player gets 6 cards. Find the probabilities of a player being dealt hands with the given characteristics.

23. All 6 cards are rooms.
24. 5 cards are suspects (the sixth can be anything).
25. None of the cards are rooms.
26. None of the cards are suspects.
27. 3 cards are suspects and 3 are weapons.
28. There are 2 cards of each type.
29. There are 3 rooms, 2 suspects, and 1 weapon.
30. There are 4 rooms and 5 suspects.

7.7 What Are the Odds?



Figure 7.33 Scratch-off lottery tickets, as well as many other games, represent the likelihood of winning using odds. (credit: "My Scratch-off Winnings" by Shoshanah/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Compute odds.
2. Determine odds from probabilities.
3. Determine probabilities from odds.

A particular lottery instant-win game has 2 million tickets available. Of those, 500,000 win a prize. If there are 500,000 winners, then it follows that there are 1,500,000 losing tickets. When we evaluate the risk associated with a game like this, it can be useful to compare the number of ways to win the game to the number of ways to lose. In the case of this game, we would compare the 500,000 wins to the 1,500,000 losses. In other words, there are 3 losing tickets for every winning ticket. Comparisons of this type are the focus of this section.

Computing Odds

The ratio of the number of equally likely outcomes in an event E to the number of equally likely outcomes *not* in the event E' is called the **odds for** (or **odds in favor of**) the event. The opposite ratio (the number of outcomes not in the event to the number in the event E' to the number in the event E is called the **odds against** the event.

⚠ Both odds and probabilities are calculated as ratios. To avoid confusion, we will always use fractions, decimals, or percents for probabilities, and we'll use colons to indicate odds. The rules for simplifying fractions apply to odds, too. Thus, the odds for winning a prize in the game described in the section opener are $500,000:1,500,000 = 1:3$ and the odds against winning a prize are $3:1$. These would often be described in words as “the odds of winning are one to three in favor” or “the odds of winning are three to one against.”

⚠ Notice that, while probabilities must always be between zero and one inclusive, odds can be any (non-negative) number, as we'll see in the next example.

EXAMPLE 7.25

Computing Odds

1. If you roll a fair 6-sided die, what are the odds for rolling a 5 or higher?
2. If you roll two fair 6-sided dice, what are the odds against rolling a sum of 7?
3. If you draw a card at random from a standard deck, what are the odds for drawing a ♡?
4. If you draw 2 cards at random from a standard deck, what are the odds against them both being ♠?

✓ Solution

1. The sample space for this experiment is $\{1, 2, 3, 4, 5, 6\}$. Two of those outcomes are in the event “roll a five or higher,” while four are not. So, the odds for rolling a five or higher are $2:4 = 1:2$.
2. In [Example 7.18](#), we found the sample space for this experiment using the following table ([Figure 7.34](#)):

		First Die					
		1	2	3	4	5	6
Second Die	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

Figure 7.34

There are 6 outcomes in the event “roll a sum of 7,” and there are 30 outcomes not in the event. So, the odds against rolling a 7 are $30:6 = 5:1$.

3. There are 13 ♡ in a standard deck, and $52 - 13 = 39$ others. So, the odds in favor of drawing a ♡ are $13 : 39 = 1:3$.
4. There are ${}_{13}C_2 = 78$ ways to draw 2 ♠, and ${}_{52}C_2 - 78 = 1,248$ ways to draw 2 cards that are *not* both ♠. So, the odds against drawing 2 ♠ are $1,248:78 = 16:1$.

> YOUR TURN 7.25

You roll a pair of 4-sided dice with faces labeled 1 through 4.

1. What are the odds for rolling a sum greater than 3?
2. What are the odds against both dice giving the same number?

Odds as a Ratio of Probabilities

We can also think of odds as a ratio of probabilities. Consider again the instant-win game from the [section opener](#), with 500,000 winning tickets out of 2,000,000 total tickets. If a player buys one ticket, the probability of winning is $\frac{500,000}{2,000,000} = \frac{1}{4}$, and the probability of losing is $1 - \frac{1}{4} = \frac{3}{4}$. Notice that the ratio of the probability of winning to the probability of losing is $\frac{1}{4} : \frac{3}{4} = 1:3$, which matches the odds in favor of winning.

FORMULA

For an event E ,

$$\begin{aligned}\text{odds for } E &= n(E):n(E') = P(E):P(E') = P(E):(1 - P(E)) \\ \text{odds against } E &= n(E'):n(E) = P(E'):P(E) = (1 - P(E)):P(E)\end{aligned}$$

We can use these formulas to convert probabilities to odds, and vice versa.

EXAMPLE 7.26

Converting Probabilities to Odds

Given the following probabilities of an event, find the corresponding odds for and odds against that event.

1. $P(E) = \frac{3}{5}$
2. $P(E) = 17\%$

Solution

1. Using the [formula](#), we have:

$$\begin{aligned}\text{odds for } E &= P(E):(1 - P(E)) \\ &= \frac{3}{5}:(1 - \frac{3}{5}) \\ &= \frac{3}{5}:\frac{2}{5} \\ &= 3:2.\end{aligned}$$

(Note that in the last step, we simplified by multiplying both terms in the ratio by 5.)
Since the odds for E are 3:2, the odds against E must be 2:3.

2. Again, we'll use the formula:

$$\begin{aligned}\text{odds for } E &= P(E):(1 - P(E)) \\ &= 0.17:(1 - 0.17) \\ &= 0.17 : 0.83 \\ &\approx 1:4.88.\end{aligned}$$

(In the last step, we simplified by dividing both terms in the ratio by 0.17.)
It follows that the odds against E are approximately 4.88:1.

YOUR TURN 7.26

1. If the probability of an event E is 80%, find the odds for and the odds against E .

Now, let's convert odds to probabilities. Let's say the odds for an event are $A : B$. Then, using the [formula above](#), we have $A:B = P(E):(1 - P(E))$. Converting to fractions and solving for $P(E)$, we get:

$$\begin{aligned}\frac{A}{B} &= \frac{P(E)}{1-P(E)} \\ A(1-P(E)) &= B \times P(E) \\ A - A \times P(E) &= B \times P(E) \\ A &= A \times P(E) + B \times P(E) \\ A &= (A+B) \times P(E) \\ \frac{A}{A+B} &= P(E).\end{aligned}$$

Let's put this result in a formula we can use.

FORMULA

If the odds in favor of E are $A:B$, then

$$P(E) = \frac{A}{A+B}.$$

EXAMPLE 7.27

Converting Odds to Probabilities

Find $P(E)$ if E :

1. The odds of E are 2:1 in favor
2. The odds of E are 6:1 against

Solution

1. Using the formula we just found, we have $P(E) = \frac{2}{2+1} = \frac{2}{3}$.
2. If the odds against are 6:1, then the odds for are 1:6. Thus, using the formula, $P(E) = \frac{1}{1+6} = \frac{1}{7}$.

YOUR TURN 7.27

Find $P(E)$ if E :

1. The odds of E are 15:1 against
2. The odds of E are 2.5:1 in favor

 Some places, particularly state lottery websites, will use the words “odds” and “probability” interchangeably. Never assume that the word “odds” is being used correctly! Compute one of the odds/probabilities yourself to make sure you know how the word is being used!

Check Your Understanding

For the following exercises, you are rolling a 6-sided die with 3 orange faces, 2 green faces, and 1 blue face.

37. What are the odds in favor of rolling a green face?
38. What are the odds against rolling a blue face?
39. What are the odds in favor of rolling an orange face?
40. What are the odds in favor of an event with probability $\frac{3}{8}$?
41. What are the odds against an event with probability $\frac{2}{13}$?
42. What is the probability of an event with odds 9:4 against?
43. What is the probability of an event with odds 5:7 in favor?



SECTION 7.7 EXERCISES

For the following exercises, find the probabilities of events with the given odds in favor.

1. 9:4
2. 2:3
3. 2:3
4. 5:4
5. 1:50
6. 7:5
7. 1:7
8. 10:9

For the following exercises, find the probabilities of events with the given odds against.

9. 1:8
10. 2:3
11. 3:2
12. 5:4
13. 1:50
14. 7:5
15. 1:7
16. 10:9

In the following exercises, find the odds in favor of events with the given probabilities. Give your answer as a ratio of whole numbers. If neither of those two numbers is 1, also give an answer as a ratio involving both 1 and a number greater than or equal to 1 (for example, the odds 5:2 and 3:8 can be reduced to 2.5:1 and 1:2.67).

17. $\frac{2}{7}$
18. $\frac{12}{17}$
19. $\frac{8}{9}$
20. $\frac{3}{8}$
21. $\frac{9}{25}$
22. $\frac{6}{7}$
23. $\frac{10}{13}$
24. $\frac{8}{15}$

In the following exercises, find the odds against events with the given probabilities. Give your answer as a ratio of whole numbers. If neither of those two numbers is 1, also give an answer as a ratio involving both 1 and a number greater than or equal to 1 (for example, the odds 5:2 and 3:8 can be reduced to 2.5:1 and 1:2.67).

25. $\frac{2}{7}$
26. $\frac{12}{17}$
27. $\frac{8}{9}$
28. $\frac{3}{8}$
29. $\frac{9}{25}$
30. $\frac{6}{7}$
31. $\frac{10}{13}$
32. $\frac{8}{15}$

In the following exercises, you are drawing from a deck containing only these 10 cards:

$A\heartsuit, A\spadesuit, A\clubsuit, A\diamondsuit, K\spadesuit, K\clubsuit, Q\heartsuit, Q\spadesuit, J\heartsuit, J\spadesuit$.

33. Let E be the event "draw an ace."
 - a. What is the probability of E ?
 - b. What are the odds in favor of E ?
 - c. What are the odds against E ?
34. Let F be the event "draw a $\heartsuit$."
 - a. What is the probability of F ?
 - b. What are the odds in favor of F ?
 - c. What are the odds against F ?
35. Let T be the event "draw two $\spadesuit$ (without replacement)."
 - a. What is the probability of T ?